

Vaddi Charan Sai Nandan Reddy

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EDUCATION

PES University

B.Tech CSE CGPA: 9.09/10

Bengaluru, Karnataka

Sep 2022 – May 2026

EXPERIENCE

Microsoft - M365 Research

Jan 2026 – July 2026

Research Intern

Bengaluru, Karnataka

- Exploring various LLM inference and serving systems including Co-located, Disaggregated Serving, Prefill-Decode Multiplexing, analyzing latency-throughput trade-offs across workloads.
- Conducted Profiling and workload-level analysis to identify bottlenecks in inference systems.
- Investigated Data and Model parallelism strategies (Tensor, Expert, Pipeline) for scaling inference.
- Explored Large-scale inference simulators to model system behaviour and improving fidelity across under-explored serving systems.

IBM

June 2025 - July 2025

GRM Mentee

Bengaluru, Karnataka

- Enabled AI inferencing on IBM z/OS by porting llama.cpp, a high-performance LLM runtime, and minja, a minimalist prompt templating engine.
- Adapted llama.cpp for z/OS by resolving platform-specific issues: memory alignment, file I/O, threading constraints, SSL CA bundle validation, and endianness-related issues.
- Improved inference stability by integrating test infrastructure and increasing pass rate to 80%.
- Explored CPU-level optimizations using SIMD and IBM MASS to improve inference performance.

PUBLICATIONS

- **Laghu: Localization of RL-Induced Reasoning in Small Language Models** - Accepted at FLINS-ISKE 2026
- **BioFlux: A Multi-Agent Reinforcement Learning Framework for Climate-Driven Ecosystem Simulation** - Best Poster Presentation at National Conference on Emerging Trends and Technologies, DSCE

PROJECTS

Mechanistic Interpretability Overview of Post-trained LLMs

Jan 2026 - April 2026

- Conducted mechanistic interpretability analysis on post-trained language models to examine the effects of post-training on reasoning and internal representations.
- Extracted and visualized attention heads, MLP activations, and residual stream compositions using TransformerLens to trace circuit-level behavior.
- Identified emergent patterns in reasoning pathways and compared pre-trained vs. post-trained model activations to interpret layer-wise functional specialization.
- Developed reusable Python scripts for automated probe training, activation collection, and neuron-level feature attribution, streamlining model interpretability experiments.

Math Agent for Diverse Contextual Math Word Problems

Aug 2025 - Dec 2025

- Developed a reinforcement learning-based math agent designed to solve contextual and multi-step math word problems beyond rote memorization.
- Implemented a policy model fine-tuned with LoRA and verified through reinforcement learning to optimize reasoning correctness across diverse problem types.
- Integrated zero- and few-shot chain-of-thought reasoning to contextualize problems.

Serverless Function Execution Engine

Jan 2025 - Feb 2025

- Developed a lightweight serverless platform enabling execution of short-lived user functions with support for Python and JavaScript.
- Implemented real-time CPU and memory monitoring and exposed metrics via REST APIs for performance tracking.
- Integrated Docker and gVisor to provide secure and isolated function execution environments.
- Optimized user experience with cold-start container reuse.

Classifier-Guided Stable Diffusion Pipeline for Image Quality Control

Jan 2025 - Apr 2025

- Proposed a classifier-guided rollback to steer the diffusion process by rejecting low-quality latent states.
- Trained a latent classifier on NIMA and CLIP-based similarity scores to predict quality and semantic alignment mid-generation.
- Implemented a Seq2Seq prompt re-writer for negative prompt optimization and increased diversity in generated outputs.
- Performed model evaluation using metrics such as ROC AUC and Brier score to assess classification and latent quality.

TECHNICAL SKILLS

Languages: Python, Go, C/C++

Frameworks: JAX, PyTorch, TensorFlow

Libraries: Hugging Face Transformers, Datasets, PeFT, TRL, reasoning-gym, DeepSpeed, LangGraph, Gymnasium

Tools: llama.cpp, vLLM, SGLang, CUDA, Nsight Systems, Kubernetes, OpenShift

Research Interests: Machine Learning, LLMs, ML Systems, LLM Serving Systems

ACHIEVEMENTS

- Open-source contributor to **zopencommunity - llama.cpp port, minja port** The Linux Foundation
- **IBM Environmental Intelligence Hackathon** — Placed Top 3
- **5× MRD Scholarship** — awarded to top 15% of semester scorers